HR ANALYTICS DASHBOARD

Employee Attrition Analysis Using Looker Studio

PROJECT REPORT: Google Looker Studio

Prepared by: Raja Kumar

Dashboard Report Link: https://datastudio.google.com/reporting/2df76bc0-4da3-4da9-9f36-22c517789679

1. Executive Summary

This project focuses on building an interactive HR Analytics Dashboard to understand employee attrition and explore how departures vary across departments and employee groups. The supplied dataset contains 1,470 employee records and 35 columns. The dashboard is designed as two pages: an HR Overview page and an Advanced Attrition Analysis page. The overview page presents key workforce KPIs and descriptive charts, while the advanced page compares attrition rates by department, age group, income bracket, job satisfaction, and gender.

The dataset contains 237 employees marked as having left and 1,233 marked as having stayed, giving an overall attrition rate of 16.12%. Sales has the highest department-level attrition rate in this dataset (about 20.63%), while Research & Development has the highest departure count (133). These results show why both counts and rates should be considered when prioritising retention investigations.

2. Project Objectives

- Summarise the workforce using KPI scorecards.
- Measure employee attrition and compare employees who left with those who stayed.
- Explore attrition by department, job role, gender, business travel, overtime, age, income, and job satisfaction.
- Create interactive filters so users can explore employee groups.
- Present findings clearly to support HR discussions and further investigation.

3. Tools and Technologies

Tool	Purpose
Microsoft Excel	Review the supplied workbook and check data quality.
Google Sheets	Host the worksheet for connection to Looker Studio.
Google Looker Studio	Build scorecards, charts, filters, calculated fields, and report pages.

The supplied project guide recommends using Google Sheets as the data source and building a two-page report. Share the report only with people authorised to view the data.

4. Dataset Overview

Item	Value
Employee records	1,470
Columns in supplied workbook	35
Employees who left (Attrition = Yes)	237
Employees who stayed (Attrition = No)	1,233
Overall attrition rate	16.12%
Average age	36.92 years
Average monthly income	6,502.93 dataset units

Key fields include Age, Attrition, Department, Gender, JobRole, MonthlyIncome, OverTime, JobSatisfaction, WorkLifeBalance, YearsAtCompany, BusinessTravel, and EmployeeNumber. The guide refers to 32 fields, but the supplied workbook contains 35 columns; the actual workbook should be treated as the source of truth.

5. Step-by-Step Project Methodology

Step 1 — Inspect the dataset

I started by opening HR Analytics Final Dataset.xlsx in Excel and checking the worksheet, column headings, and employee records. I reviewed important fields such as Attrition, Department, Age, MonthlyIncome, and EmployeeNumber.

- Check for blank or missing values in important fields.
- Confirm Age, MonthlyIncome, YearsAtCompany, and EmployeeNumber are numeric.
- Check that Attrition and OverTime use consistent Yes/No values.
- Check whether EmployeeNumber uniquely identifies each employee.
- Keep the original workbook unchanged and work from a copy.

Step 2 — Import the workbook into Google Sheets

1. Open Google Sheets and create a new spreadsheet.
2. Choose File → Import → Upload and select HR Analytics Final Dataset.xlsx.
3. Import the data into the new spreadsheet.
4. Rename the worksheet to HR Data.
5. Confirm that the expected records and columns have been imported.

Step 3 — Connect the data to Looker Studio

6. Open Looker Studio and create a new report.
7. Choose the Google Sheets connector.
8. Select the spreadsheet and the HR Data worksheet.
9. Confirm that the first row is used as column headers.
10. Add the data source and check field names and data types.

Step 4 — Build Page 1: HR Overview Dashboard

I created the first page to give HR users a quick view of the workforce. It includes four KPI scorecards, five charts, and interactive filters.

Dashboard element	Configuration / purpose
KPI: Total Employees	Distinct count of EmployeeNumber; expected 1,470.
KPI: Employees Who Left	Count employees where Attrition = Yes; expected 237.
KPI: Average Monthly Income	Average of MonthlyIncome; expected 6,502.93.
KPI: Average Employee Age	Average of Age; expected 36.92.
Attrition split	Pie or donut chart comparing Yes and No.
Attrition by Department	Bar chart filtered to Attrition = Yes.
Employee Count by Job Role	Horizontal bar chart by JobRole.
Gender Distribution	Pie chart by Gender.
Attrition by Business Travel	100% stacked bar chart by BusinessTravel and Attrition.
Filters	Department, Gender, and OverTime.

I placed KPI cards at the top and charts below them. I used clear titles, consistent colours, readable labels, and enough spacing to keep the report easy to follow.

Step 5 — Create Attrition Flag

To calculate attrition numerically, I created a calculated field named Attrition Flag. It assigns 1 when Attrition is Yes and 0 otherwise.

```
CASE
  WHEN Attrition = "Yes" THEN 1
  ELSE 0
END
```

Validation: the sum of Attrition Flag should equal 237.

Step 6 — Create Attrition Rate

I created an Attrition Rate field by dividing the sum of Attrition Flag by the distinct employee count. The result is formatted as a percentage.

```
SUM(Attrition Flag) / COUNT_DISTINCT(EmployeeNumber)
```

For the overall dataset, 237 / 1,470 = 16.12%. When used in charts by department or another group, the numerator and denominator must be evaluated within the same group. Validate the results in Looker Studio before relying on them.

Step 7 — Create employee grouping fields

I added three descriptive fields to make the advanced analysis easier to interpret.

Calculated field	Grouping logic	Purpose
Age Group	Under 25; 25–34; 35–44; 45 and above	Compare attrition across age bands.
Income Bracket	Below 3,000; 3,000–5,999; 6,000–9,999; 10,000 and above	Compare attrition across income ranges.
Satisfaction Level	1-Low; 2-Medium; 3-High; 4-Very High	Compare attrition by job satisfaction score.

The exact formula conditions determine which group receives employees whose values fall on a boundary. Check the calculated-field editor and validate the group labels.

Step 8 — Build Page 2: Advanced Attrition Analysis

The second page uses calculated fields to compare attrition rates rather than only showing raw departure counts.

Visual	Dimension(s)	Metric / purpose
Attrition Rate by Department	Department	Compare department attrition rates.
Attrition Rate by Age Group	Age Group	Compare age bands.
Attrition Rate by Income Bracket	Income Bracket	Compare income groups.
Attrition Rate by Satisfaction Level	Satisfaction Level	Compare satisfaction groups.
Department and Gender Pivot Table	Department rows; Gender columns	Compare rates side by side.
Filters	JobRole, Age Group, Income Bracket, OverTime	Enable focused analysis.

I used percentage formatting for attrition-rate charts and meaningful chart titles. I checked that age, income, and satisfaction categories appeared in a logical order.

Step 9 — Validate and share the report

- Check that the employee count is 1,470 and the departure count is 237.
- Confirm that the overall attrition rate is about 16.12%.
- Check average age and average monthly income against the dataset.
- Test each filter and confirm relevant charts update.
- Review chart titles, labels, number formats, and navigation.
- Set sharing permissions carefully and do not publish employee data without authorization.

6. Key Findings

Finding	Result	Interpretation
Overall attrition	237 of 1,470 employees; 16.12%	About one in six employees in this dataset is marked as having left.
Sales department	92 departures; approx. 20.63% attrition	Sales has the highest department-level attrition rate among the three departments.
Research & Development	133 departures; approx. 13.84% attrition	R&D has the largest number of departures, but not the highest rate.
Human Resources	12 departures; approx. 19.05% attrition	Interpret the rate in the context of the department's smaller employee count.

These findings are descriptive. They show where attrition is concentrated in this dataset, but they do not prove why employees left. HR should combine the dashboard with exit interviews, engagement feedback, and other information before deciding on interventions.

7. Recommendations

- Review reasons for elevated attrition in Sales using exit interviews and employee feedback.
- Investigate the large number of departures in Research & Development while considering its larger workforce.
- Compare attrition rates by overtime, satisfaction, age, and income to identify groups that may need further review.
- Use consistent definitions for employee counts and attrition rates across all charts.
- Refresh the source data periodically if the dashboard is intended for ongoing monitoring.
- Do not treat correlation or group differences as proof of causation.

8. Conclusion

This project demonstrates how an employee dataset can be turned into an interactive HR dashboard using Google Sheets and Looker Studio. The overview page summarises workforce size and attrition, while the advanced page compares attrition rates across employee groups. Sales has the highest department-level attrition rate, whereas Research & Development has the highest number of departures. The dashboard provides a useful starting point for HR investigations and retention planning, provided its metrics are validated and interpreted alongside additional business context.

9. Portfolio Summary

Built a two-page HR Attrition Analytics Dashboard in Google Looker Studio using a 1,470-record employee dataset. Prepared data in Google Sheets, developed KPI scorecards and interactive charts, created calculated fields for attrition rate and employee segmentation, and identified department-level differences in attrition to support further HR investigation.